

Transformation of Variables

1 Model Inadequacies and Transformations

Correcting Model Inadequacies

- Least-squares inference assumes a linear relationship, constant error variance, and normally distributed errors
- Residual plots reveal when these assumptions fail
- One standard remedy is to transform the data; weighting is a separate topic
- In this lecture: transforming y to stabilize the variance, transforming y or x to linearize the model, and two analytical methods for choosing the power (Box–Cox for y , Box–Tidwell for x)

Recognizing Nonconstant Variance

- Plot R-student against the fitted values $\hat{y}_i$
- Outward-opening funnel: the spread grows with $E(y_i)$, common when y is a count or an amount
- Double bow: the spread is largest in the middle and small at the extremes, typical when y is a proportion
- With nonconstant variance the OLS estimators remain unbiased but are no longer minimum-variance, and the usual t and F tests are unreliable

Idea of a Variance-Stabilizing Transformation

- Suppose $\text{Var}(y_i)$ depends on the mean $E(y_i)$
- Replace y by $y^* = h(y)$, with h chosen so that $\text{Var}(y^*)$ is approximately constant
- Then fit the model on the transformed scale

$$y_i^* = \beta_0 + \beta_1 x_i + \epsilon_i, \quad \text{Var}(\epsilon_i) \approx \sigma^2$$

- The appropriate h depends on how $\text{Var}(y)$ is related to $E(y)$

2 Variance-Stabilizing Transformations

Variance-Stabilizing Transformations

Relationship of σ^2 to $E(y)$	Transformation
$\sigma^2 \propto \text{constant}$	$y^* = y$ (no transformation)
$\sigma^2 \propto E(y)$	$y^* = \sqrt{y}$ (Poisson data)
$\sigma^2 \propto E(y)[1 - E(y)]$	$y^* = \sin^{-1}(\sqrt{y})$ (binomial proportions, $0 \leq y_i \leq 1$)
$\sigma^2 \propto [E(y)]^2$	$y^* = \ln y$
$\sigma^2 \propto [E(y)]^3$	$y^* = y^{-1/2}$
$\sigma^2 \propto [E(y)]^4$	$y^* = y^{-1}$

Reading the Table

- The stronger the dependence of σ^2 on $E(y)$, the stronger the transformation: $\sqrt{y}$, then $\ln y$, then $y^{-1/2}$, then y^{-1}
- Poisson counts have $\text{Var}(y) = E(y)$, so $\sigma^2 \propto E(y)$ and the square root applies
- A proportion $y = X/m$ with $X \sim \text{Binomial}(m, p)$ has $\text{Var}(y) = p(1 - p)/m$, so $\sigma^2 \propto E(y)[1 - E(y)]$ and the arcsine square root applies
- $\sigma^2 \propto [E(y)]^2$ means the standard deviation is proportional to the mean, which calls for the logarithm

Choosing a Transformation in Practice

1. Fit the model by OLS on the original y and plot R-student against $\hat{y}_i$
2. If the pattern of spread matches a row of the table, apply that transformation
3. Refit with y^* and examine the residual plots again
4. If the variance is still not constant, try a stronger or weaker transformation

The exact relationship between σ^2 and $E(y)$ is rarely known, so the choice is often made by trial and error.

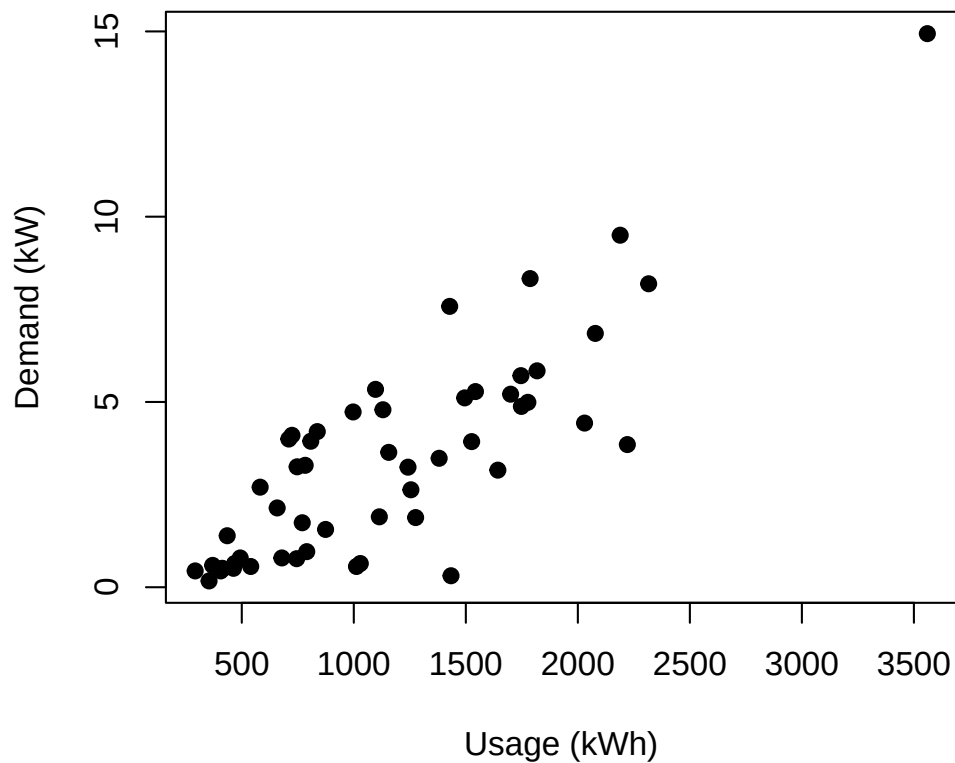
Consequences of Transforming the Response

- Estimates and tests refer to the y^* scale; for example, coefficients describe the change in $\sqrt{y}$, not in y
- To predict on the original scale, apply the inverse transformation to fitted values and to interval endpoints
- R^2 values from models with different responses cannot be compared
- Transforming y also changes the shape of the relationship and the error distribution, so all residual checks must be repeated

3 Example: Electric Utility Data

Electric Utility Data

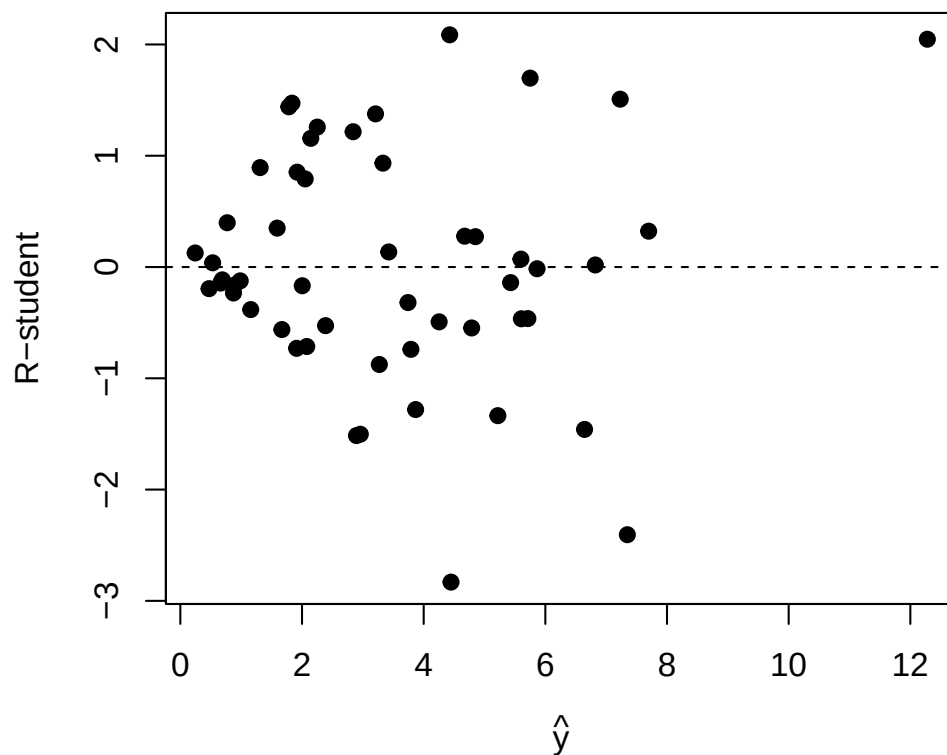
- y : peak demand (kW), x : monthly energy usage (kWh)
- $n = 53$ residential customers in August (MPV Table 5.2)
- Goal: model demand as a function of usage
- The spread of y increases with x



Straight-Line Fit

$$\hat{y} = -0.8313 + 0.003683x$$
$$R^2 = 0.7046$$

- R-student against $\hat{y}$ shows an outward-opening funnel
- The constant-variance assumption is violated



Question: Which Transformation?

The residual plot is an outward-opening funnel, so the variance increases with the mean. Which row of the variance-stabilizing table would you try first, and what would you do if the funnel persisted?

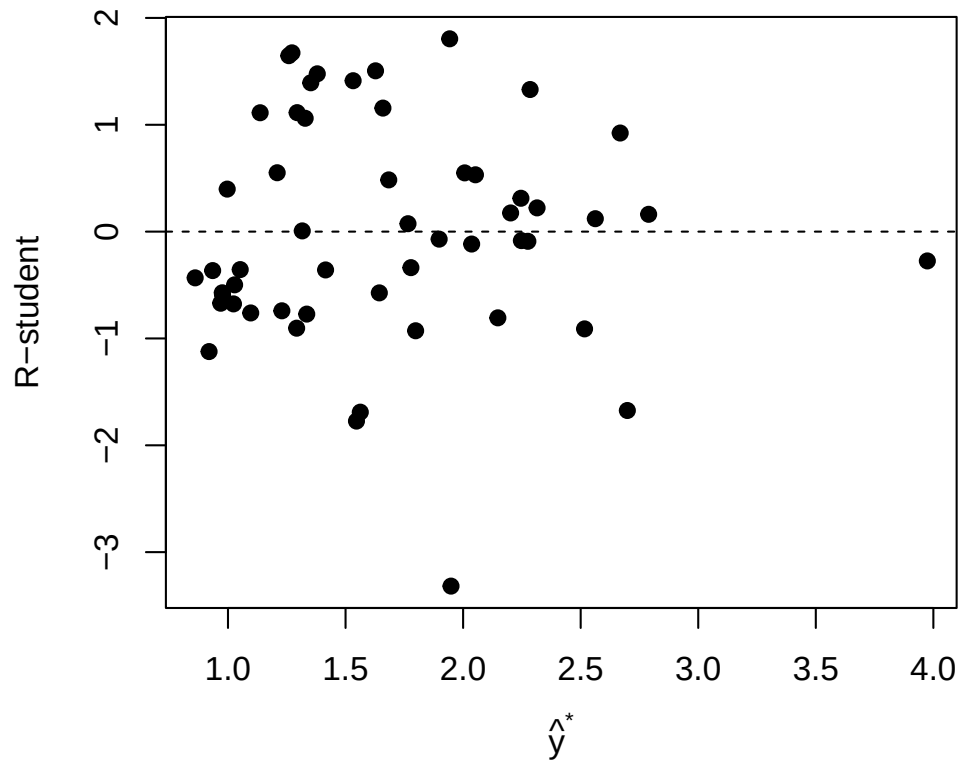
Square-Root Transformation

$$y^* = \sqrt{y}$$

$$\hat{y}^* = 0.5822 + 0.0009529 x$$

$$R^2 = 0.6485$$

- R-student against $\hat{y}^*$ no longer shows a funnel
- The variance now looks stable



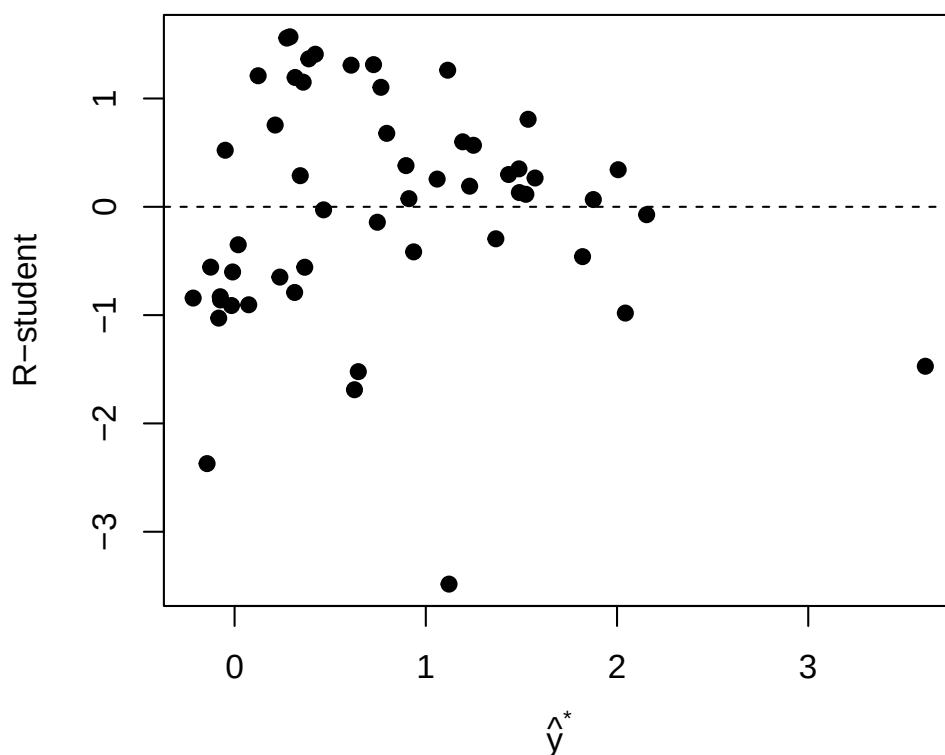
A Stronger Transformation: $\ln y$

$$y^* = \ln y$$

$$\hat{y}^* = -0.5587 + 0.001172 x$$

$$R^2 = 0.5266$$

- The residual plot now shows curvature
- The logarithm is too strong for these data



Question: Did R^2 Get Worse?

R^2 dropped from 0.7046 for y to 0.6485 for $\sqrt{y}$. Does this mean the transformed model is worse?

4 Transformations to Linearize the Model

Linearizable Nonlinear Functions

Function	Transformation	Linear form
$y = \beta_0 x^{\beta_1}$	$y' = \log y, x' = \log x$	$y' = \log \beta_0 + \beta_1 x'$
$y = \beta_0 e^{\beta_1 x}$	$y' = \ln y$	$y' = \ln \beta_0 + \beta_1 x$
$y = \beta_0 + \beta_1 \log x$	$x' = \log x$	$y = \beta_0 + \beta_1 x'$
$y = \frac{x}{\beta_0 x - \beta_1}$	$y' = \frac{1}{y}, x' = \frac{1}{x}$	$y' = \beta_0 - \beta_1 x'$

The Error Structure Matters

- Linearizing the exponential function assumes the error is multiplicative

$$y = \beta_0 e^{\beta_1 x} \epsilon \implies \ln y = \ln \beta_0 + \beta_1 x + \ln \epsilon$$

- The linearized model is appropriate if $\ln \epsilon$ has mean zero, constant variance, and is normal
- If instead the error is additive, $y = \beta_0 e^{\beta_1 x} + \epsilon$, taking logarithms does not give a linear model with additive error, and nonlinear regression is needed

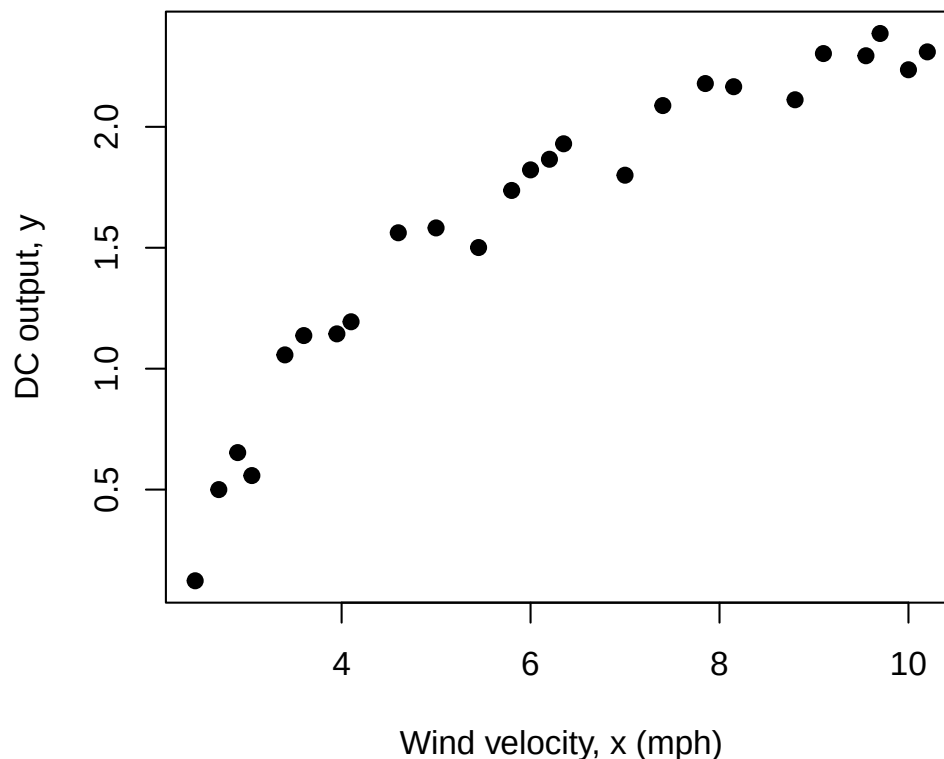
Transform x or Transform y ?

- Transforming x alone leaves the distribution of the errors unchanged
- Transforming y changes both the shape of the relationship and the error distribution
- Curved scatter with roughly constant variance: transform x
- Curved scatter with nonconstant variance: transform y , which may fix both problems

5 Example: Windmill Data

Windmill Data

- y : DC output of a windmill, x : wind velocity (mph)
- $n = 25$ observations (MPV Table 5.5)
- Goal: model DC output as a function of wind velocity
- The scatter plot is clearly curved



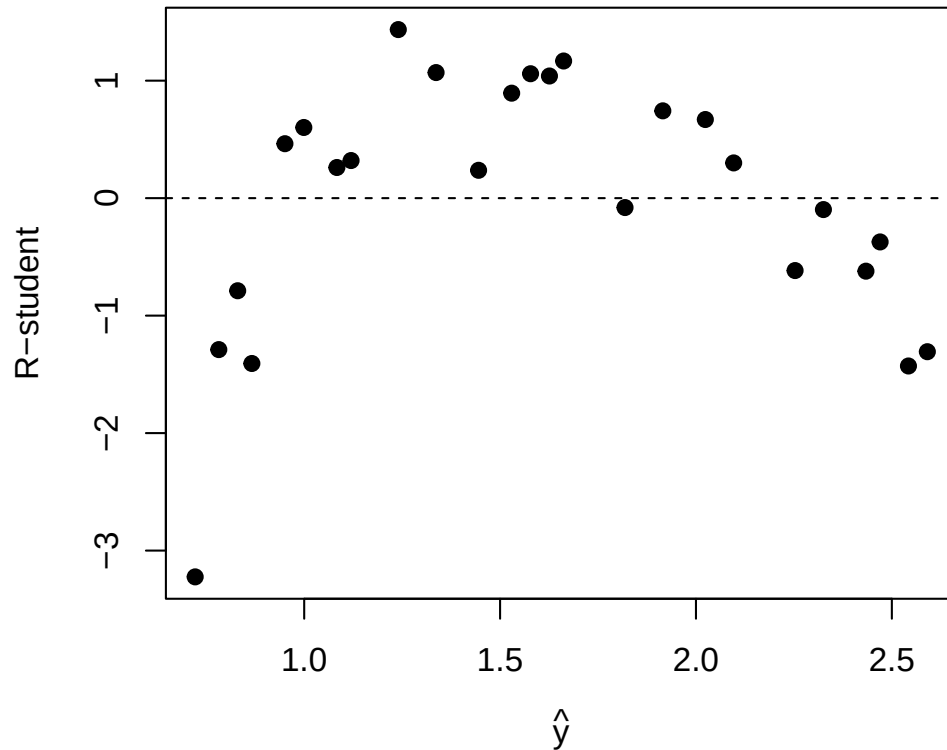
Question: Which Transformation?

DC output rises with wind velocity but levels off, and the spread around the trend looks roughly constant. Which transformation would you try, and on which variable?

Straight-Line Fit

$$\hat{y} = 0.1309 + 0.2411 x$$
$$R^2 = 0.8745$$

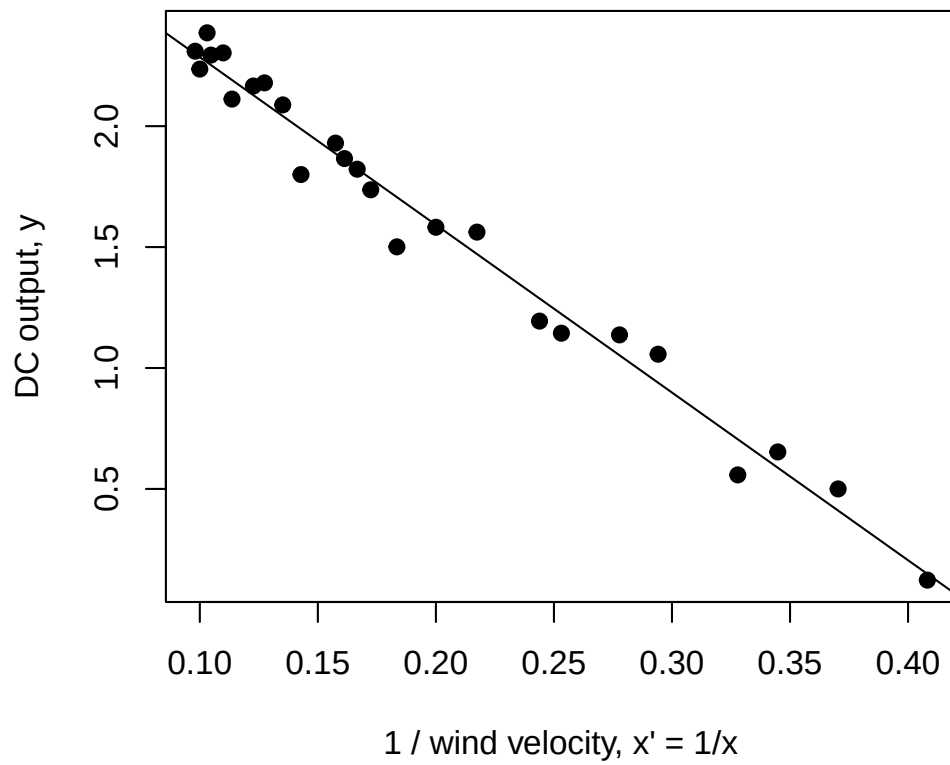
- R-student against $\hat{y}$ shows a clear curved pattern
- The straight line is inadequate despite the high R^2



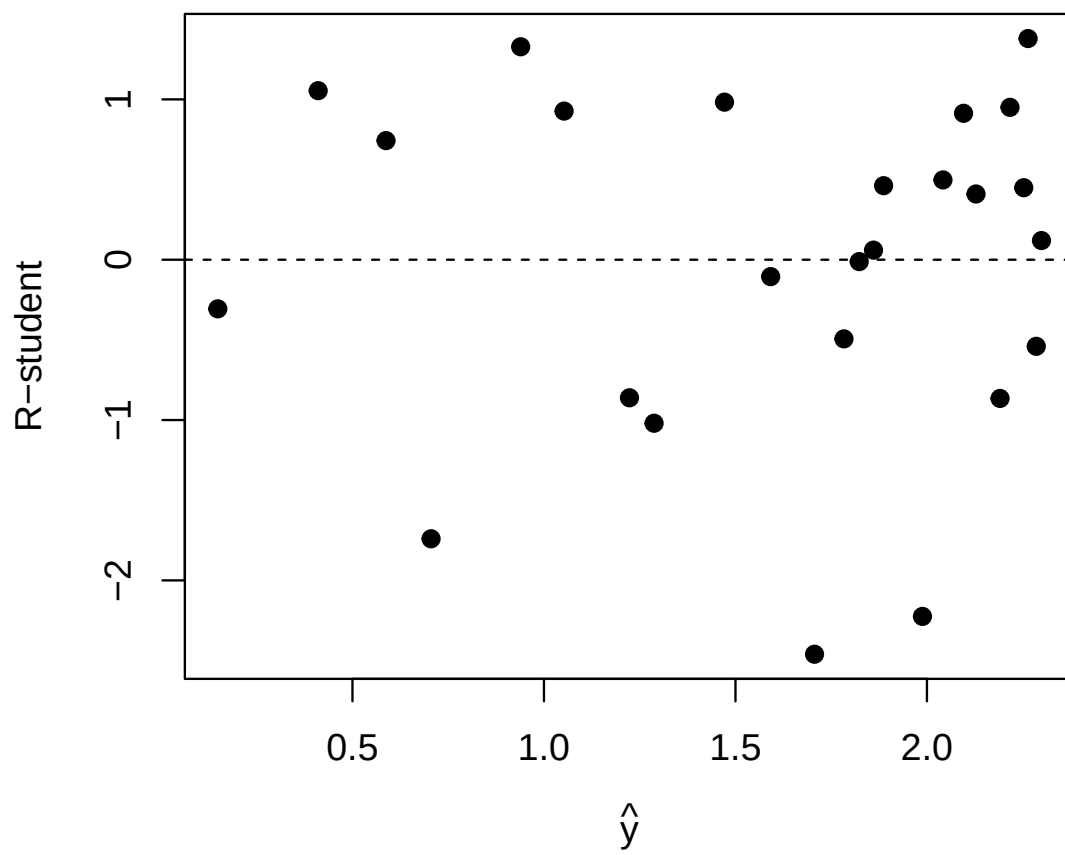
Reciprocal Transformation of x

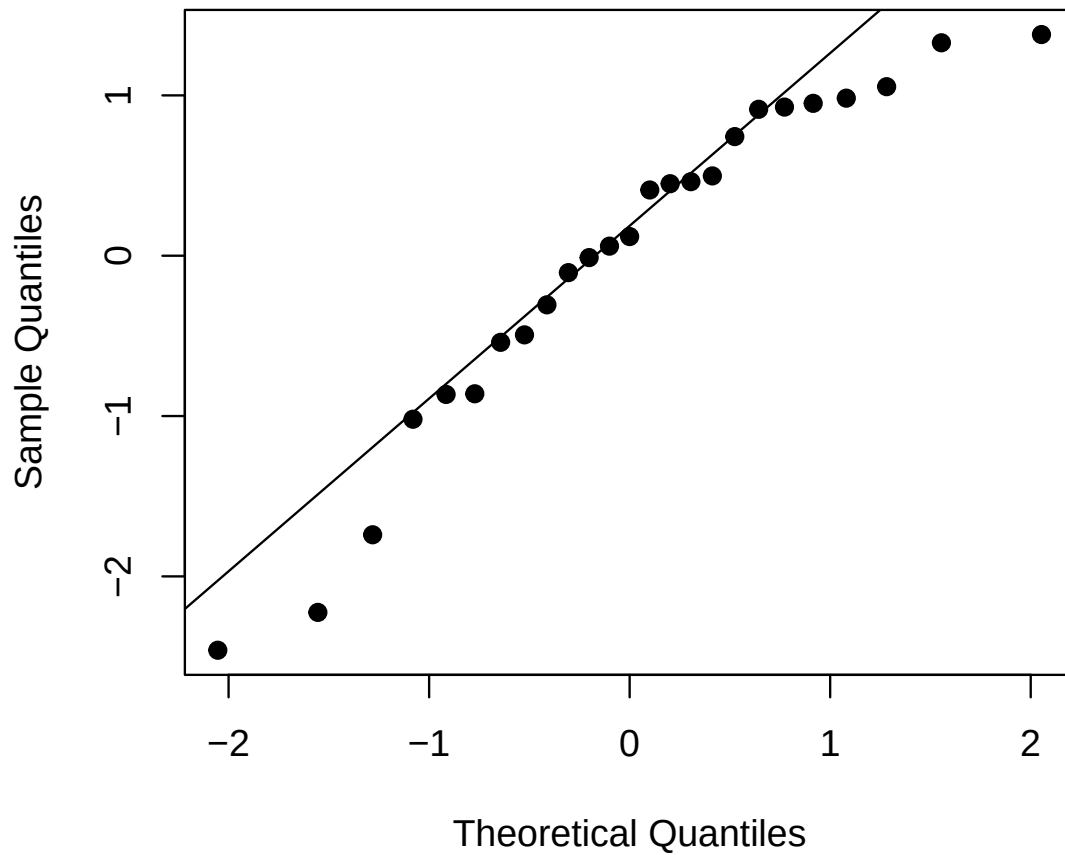
$$x' = 1/x$$
$$\hat{y} = 2.9789 - 6.9345 x'$$
$$R^2 = 0.9800$$

- The scatter plot against x' is close to a straight line
- As $x \rightarrow \infty$ the fitted output approaches 2.9789, the upper limit



Residual Analysis After the Transformation





- R-student against $\hat{y}$ shows no systematic pattern
- The normal probability plot is close to a straight line

6 The Box–Cox Method

Motivation for the Box–Cox Method

- The variance-stabilizing table requires knowing how σ^2 depends on $E(y)$, which is usually unknown
- Box and Cox (1964): estimate the power of the transformation from the data
- Considers the power family y^λ , with λ estimated together with β by maximum likelihood
- Aims to correct nonnormality and nonconstant variance, and requires $y_i > 0$

The Power Transformation Family

$$y^{(\lambda)} = \begin{cases} \frac{y^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \ln y, & \lambda = 0 \end{cases}$$

- As $\lambda \rightarrow 0$, $(y^\lambda - 1)/\lambda \rightarrow \ln y$, so the logarithm belongs to the family
- $\lambda = 1$: no transformation, $\lambda = 1/2$: square root, $\lambda = -1$: reciprocal

The Scaled Response

- The size of y^λ changes with λ , so SS_{Res} is not comparable across values of λ
- Fix: divide by a factor involving the geometric mean $\dot{y}$

$$y^{(\lambda)} = \begin{cases} \frac{y^\lambda - 1}{\lambda \dot{y}^{\lambda-1}}, & \lambda \neq 0 \\ \dot{y} \ln y, & \lambda = 0 \end{cases} \quad \dot{y} = \exp \left[\frac{1}{n} \sum_{i=1}^n \ln y_i \right]$$

Model and Estimation

$$\mathbf{y}^{(\lambda)} = \mathbf{X}\beta + \epsilon, \quad \epsilon \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$$

- For fixed λ , $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}^{(\lambda)}$
- Let $SS_{\text{Res}}(\lambda)$ be the residual sum of squares from this fit

$$SS_{\text{Res}}(\lambda) = \sum_{i=1}^n (y_i^{(\lambda)} - \hat{y}_i^{(\lambda)})^2$$

- With the scaled response, the maximum likelihood estimate $\hat{\lambda}$ is the value that minimizes $SS_{\text{Res}}(\lambda)$

The Box–Cox Procedure

1. Choose a grid of values of λ , for example from -2 to 2
2. For each λ , compute the scaled $y^{(\lambda)}$, fit the model, and record $SS_{\text{Res}}(\lambda)$
3. Plot $SS_{\text{Res}}(\lambda)$ against λ and locate the minimum $\hat{\lambda}$
4. Analyze the simple power y^λ (not the scaled version), using a convenient value of λ near $\hat{\lambda}$, and recheck the residuals

Later inference treats λ as known, so it is only approximate.

Confidence Interval for λ

$$SS^* = SS_{\text{Res}}(\hat{\lambda}) \left(1 + \frac{t_{\alpha/2, \nu}^2}{\nu} \right)$$

- ν is the residual degrees of freedom
- A $100(1 - \alpha)\%$ confidence interval for λ contains all λ with $SS_{\text{Res}}(\lambda) \leq SS^*$
- If the interval contains 1, no transformation is needed
- Otherwise choose a convenient value inside the interval, such as -1 , 0 , $1/2$

Log-Likelihood Form and **MASS::boxcox**

- Up to a constant, the profile log-likelihood of λ is

$$L(\lambda) = -\frac{n}{2} \ln SS_{\text{Res}}(\lambda)$$

- The likelihood-based interval keeps all λ with $L(\hat{\lambda}) - L(\lambda) \leq \frac{1}{2}\chi_{\alpha,1}^2$
- Equivalently, $SS_{\text{Res}}(\lambda) \leq SS_{\text{Res}}(\hat{\lambda}) \exp(\chi_{\alpha,1}^2/n)$, which agrees closely with SS^* unless n is small
- **MASS::boxcox** plots $L(\lambda)$ with this 95% interval

7 Example: Box–Cox on the Electric Utility Data

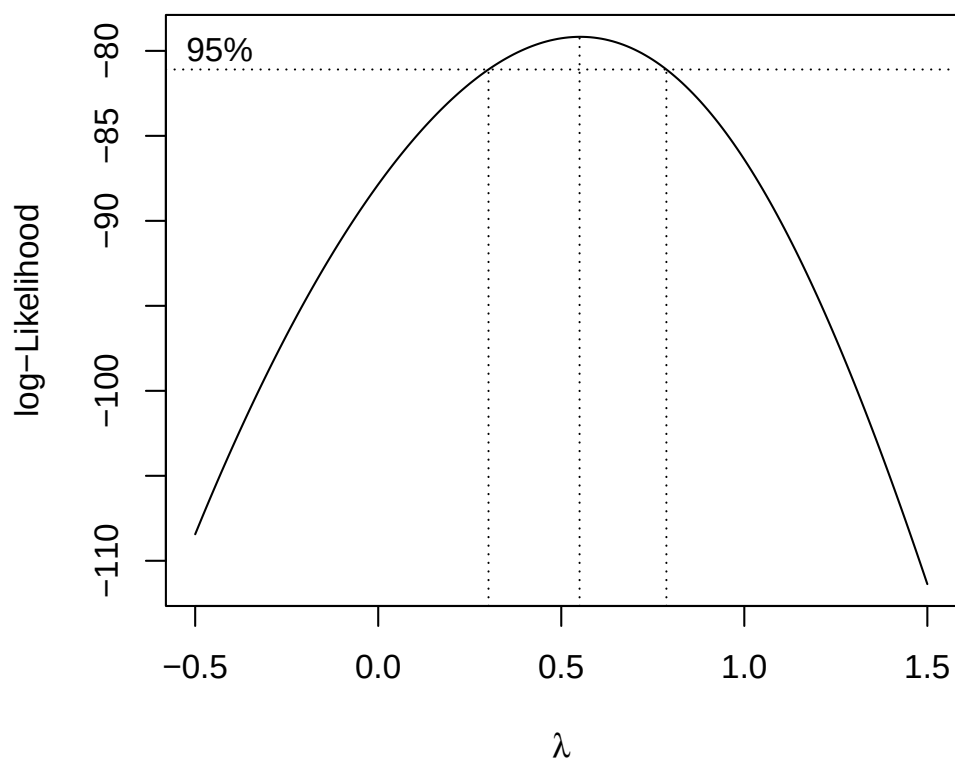
$SS_{\text{Res}}(\lambda)$ for the Electric Utility Data

λ	$SS_{\text{Res}}(\lambda)$
−1	986.03
−0.5	291.58
0	134.09
0.25	107.21
0.5	96.95
0.75	101.69
1	126.87

- The scaled $SS_{\text{Res}}(\lambda)$ is smallest near $\lambda = 0.5$

Box–Cox Plot from **MASS**

- $\hat{\lambda} = 0.55$
- 95% interval: (0.31, 0.78)
- Contains $\lambda = 0.5$
- Excludes $\lambda = 1$ and $\lambda = 0$



Question: Which λ Should We Use?

The estimate is $\hat{\lambda} = 0.55$ with 95% interval $(0.31, 0.78)$. Which value would you use for the final model, and why?

8 The Box–Tidwell Method

Motivation for the Box–Tidwell Method

- Box–Cox transforms y ; when the problem is nonlinearity with constant variance, transform the regressor instead
- Model, with unknown power α :

$$y = \beta_0 + \beta_1 \xi + \epsilon, \quad \xi = \begin{cases} x^\alpha, & \alpha \neq 0 \\ \ln x, & \alpha = 0 \end{cases}$$

- Box and Tidwell (1962) estimate α iteratively
- Special case: $\alpha = 0$: logarithm

Linearizing the Model in α

Expand $E(y) = \beta_0 + \beta_1 x^\alpha$ in a first-order Taylor series in α about $\alpha_0 = 1$, using $\partial x^\alpha / \partial \alpha = x^\alpha \ln x$:

$$\begin{aligned} E(y) &\approx \beta_0 + \beta_1 x + (\alpha - 1) \beta_1 x \ln x \\ &= \beta_0 + \beta_1 x + \gamma w, \quad \gamma = (\alpha - 1) \beta_1, \quad w = x \ln x \end{aligned}$$

The model is now linear in the regressors x and w , and $\alpha = \gamma/\beta_1 + 1$.

One Iteration of the Procedure

1. Fit $y = \beta_0 + \beta_1 x + \epsilon$ and record $\hat{\beta}_1$
2. Fit $y = \beta_0^* + \beta_1^* x + \gamma w + \epsilon$ with $w = x \ln x$ and record $\hat{\gamma}$
3. Compute the first estimate of the power

$$\hat{\alpha}_1 = \frac{\hat{\gamma}}{\hat{\beta}_1} + 1$$

Iterating

- Set $x^{(1)} = x^{\hat{\alpha}_1}$ and repeat the procedure with $x^{(1)}$ in place of x
- Stop when $\hat{\alpha}_k \approx 1$, that is, when no further transformation is needed
- The overall estimate is the product $\hat{\alpha} = \prod_k \hat{\alpha}_k$, since a power of a power multiplies the exponents
- The t statistic for $\hat{\gamma}$ gives an approximate test of $H_0: \alpha = 1$

9 Example: Box–Tidwell on the Windmill Data

Windmill Data: First Iteration

$$\hat{\beta}_1 = 0.2411, \quad \hat{\gamma} = -0.4626 \quad (t = -9.13)$$

$$\hat{\alpha}_1 = \frac{-0.4626}{0.2411} + 1 = -0.918$$

- The significant $\hat{\gamma}$ confirms that the straight line is inadequate
- $\hat{\alpha}_1$ is close to -1 , which suggests the reciprocal transformation

Windmill Data: Iterations

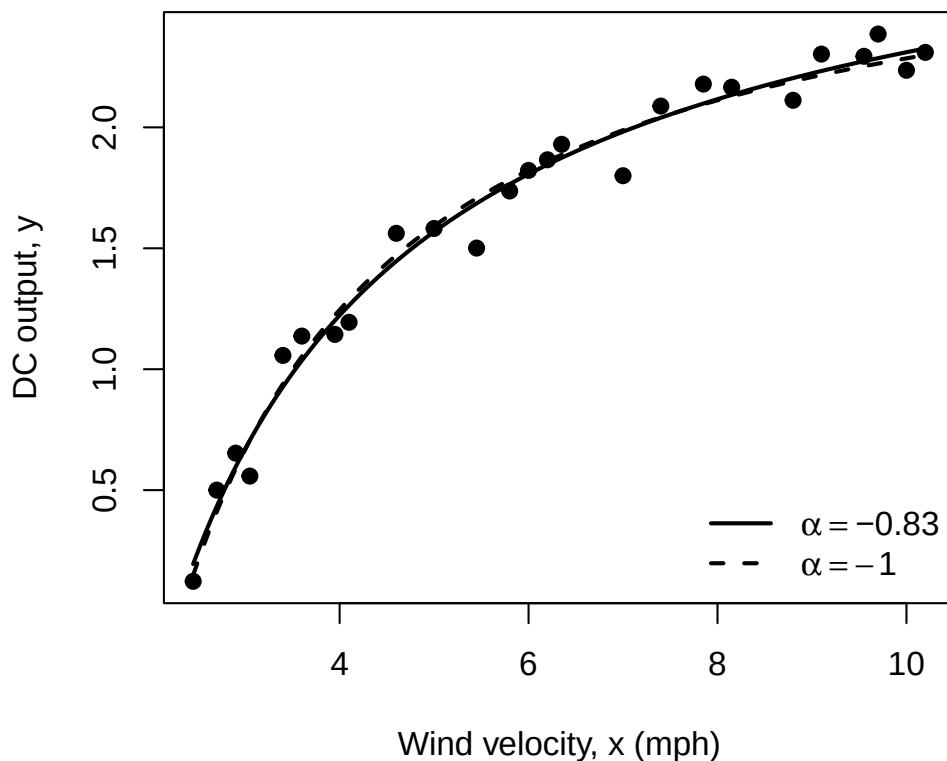
k	$\hat{\beta}_1$	$\hat{\gamma}$	$t(\hat{\gamma})$	$\hat{\alpha}_k$	cumulative $\hat{\alpha}$
1	0.2411	-0.4626	-9.13	-0.9183	-0.9183
2	-6.6784	0.5994	0.51	0.9103	-0.8359
3	-6.4738	0.0190	0.02	0.9971	-0.8334
4	-6.4686	0.0006	0.00	0.9999	-0.8333

- From the second iteration on, $\hat{\gamma}$ is not significant and $\hat{\alpha}_k \approx 1$
- The procedure converges to $\hat{\alpha} = -0.833$

Box–Tidwell Fit Compared with the Reciprocal Model

$$\hat{y} = 3.2601 - 6.4684x^{-0.833}$$

	$\alpha = -0.833$	$\alpha = -1$
R^2	0.9809	0.9800
$\hat{\sigma}$	0.0920	0.0942



Question: $x^{-0.833}$ or $1/x$?

The estimated power is -0.833 , but the reciprocal model has almost the same R^2 . Which model would you report?

10 Summary

Summary

- Nonconstant variance: transform y using the variance-stabilizing table, then recheck the residuals
- Unknown variance–mean relationship: estimate the power of y with the Box–Cox method
- Curved relationship with constant variance: transform x , guided by the shape of the scatter plot

- Unknown power for x : estimate it with the Box–Tidwell method
- After any transformation, repeat the residual analysis and interpret results on the transformed scale